

Jordan Chu

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EDUCATION

Northwestern University

Evanston, IL

B.S. in Mechanical Engineering, Design Concentration | GPA: 3.57/4.0, Dean's List

Expected June 2027

- **Kellogg School of Management:** Certificate Program for Undergraduates (CPU) in Managerial Analytics
- **Relevant Coursework:** Theory of Machines (Machine Design), Mechanics of Materials, Manufacturing & Design, Properties of Materials, Thermodynamics, Fluid Mechanics, Heat Transfer, Electronic Design, Engineering Analysis I-IV
- **Activities:** Kappa Theta Pi (Member, V.P. Recruitment), Rock-Climbing Club, Taiwanese American Student Assoc.

EXPERIENCE

GE Appliances, a Haier company

Louisville, Kentucky

Mechanical Design Engineering Intern

June 2026 – Present

- Validating spring and damper designs for a sub-washer system through analytical calculations and dyno testing.
- Coordinating with suppliers on component sourcing, sampling timelines, and design-for-manufacturing feedback.

Mechanical Design Engineering Intern

Aug. 2025 – Dec. 2025

- Redesigned the Monogram refrigerator door bin to consolidate complex geometry, cutting tooling count and unit cost.
- Partnered with an existing supplier to prototype and validate the redesign, saving an estimated \$230,000 annually.
- Developed an acrylic coupler to isolate an LED control board from water ingress, passing the UL spill test.
- Engineered an auto-open dishwasher drying feature with a hydrophobic blocker, cutting cycle energy 18%; patent filed.

Aerodynamics Engineer

Aug. 2024 – June 2025

Northwestern Formula SAE

Evanston, IL

- Designed a carbon-fiber undertray for the competition vehicle, shaping floor geometry to add downforce and cut drag.
- Ran 2D and 3D CFD simulations in Ansys Fluent, achieving a 13% increase in downforce over the prior year's car.
- Performed FEA to verify mounting integrity and manufactured the final undertray via a carbon-fiber wet layup.

Research Intern

June 2024 – Sept. 2024

Hoelzle Research Lab, The Ohio State University

Columbus, OH

- Partnered with PhD students on trilinear-coordinate scanning to enable 3D printing on curved surgical surfaces.
- Generated parametric 3D surfaces in MATLAB, CT-scanned at Wexner Medical Center; presented at 2024 SFFS.
- Converted CT data into Bezier surfaces via the Jafari-Gans method to build a virtual point cloud for printing.
- Programmed the printer to follow the point cloud; built a four-bar-linkage polycarbonate camera mount for the rig.

PROJECTS

Two-Stage Spur Gear Reducer | SolidWorks, GD&T, AGMA, LaTeX

Spring 2026

- Designed a 70 kW two-stage spur gear reducer (2600 → 299 rpm, 8.68:1 ratio) as part of a three-person team.
- Sized all three shafts with Goodman fatigue and yield analysis, confirming infinite life at a minimum FoS of 1.9.
- Ran AGMA bending and contact (pitting) analysis on all four gears for 99% reliability and infinite fatigue life.
- Selected SKF deep-groove bearings via L_{10} life calculations and set a 5-year maintenance and replacement plan.
- Produced dimensioned SolidWorks part and assembly drawings with ISO fits (k6/H7), tolerances, and a complete BOM.

Rear Bike Brake Caliper | Siemens NX, FEA, Additive Manufacturing

Winter 2025

- Applied topology optimization and FEA in Siemens NX to cut caliper mass while meeting strength/stiffness targets.
- Manufactured iterative Nylon-12 prototypes via additive manufacturing to reach a lightweight, test-ready design fast.
- Validated braking distance and maximum displacement against ISO 4210-2 and ISO 4210-4 bicycle safety standards.

Jack-in-a-Box Dynamics Simulator | Python, Lagrangian Mechanics

Fall 2025

- Simulated rigid-body impact dynamics of a jack in a box using Lagrangian mechanics and the Euler-Lagrange equations.
- Built transformation matrices in Python to express jack and box positions within a fixed world reference frame.
- Implemented per-timestep collision detection and solved nonlinear impact equations for post-impact velocities.

TECHNICAL SKILLS

CAD: SolidWorks, Siemens NX, Creo, Fusion 360

Analysis & Simulation: Ansys Fluent (CFD), FEA, AGMA Gear Design, GD&T / Tolerance Stack-up, PolyWorks

Manufacturing: Additive Manufacturing (FDM/SLA), Carbon-Fiber Layup, Design for Manufacturing (DFM)

Programming & Tools: MATLAB, Python, LaTeX, Windchill (PLM), Microsoft Office

Languages: English (Native), Mandarin (Fluent), Spanish (Intermediate)

Interests: Soccer, Running, Pickleball, Rock-Climbing, Cello